

Original MCQs

Question 363 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

The faces of a regular dodecahedron are:

- A. *pentagons.*
- B. *pentagonsandtriangles.*
- C. *trianglesandsquares.*
- D. *triangles.*

Question 364 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

If the basis of a pyramid is a n -sided polygon, how many faces does the pyramid have?

- A. It has $n + 1$ faces.
- B. It has $n + 2$ faces.
- C. It has n faces.
- D. It has $n - 1$ faces.

Question 366 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

How many faces does a prism have if the bases are n -sided polygons?

- A. It has $n + 2$ faces.
- B. It has n faces.
- C. It has $n + 3$ faces.
- D. It has $n + 1$ faces.

Question 367 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Identify the prisms that have less than 6 vertices.

- A. *Therearenosuchprisms.*
- B. *Hexagonalprisms.*
- C. *Pentagonalprisms.*
- D. *Heptagonalprisms.*

Question 368 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Identify the true statement.

- A. *Theregular tetrahedron has four geometrically equal faces.*
- B. *The tetrahedron has less than four faces.*
- C. *The tetrahedron has at least four faces.*
- D. *The tetrahedron is a platonic solid.*

Question 369 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

A quadrilateral is a rectangle when

- A. *all four internal angles are congruent.*
- B. *it has congruent internal angles two by two OR it has two pairs of congruent internal angles.*
- C. *it has congruent sides two by two OR it has two pairs of congruent sides.*
- D. *it has parallel sides two by two OR it has two pairs of parallel sides.*

Question 371 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

In a parallelogram

- A. *the two diagonals bisect.*
- B. *the two diagonals are perpendicular.*
- C. *the four sides are congruent.*
- D. *the four angles are congruent.*

Question 456 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

In a triangle, the centroid (or barycenter) is the point of intersection of

- A. *the medians of the triangle.*
- B. *the bisectors of the internal angles of the triangle.*
- C. *the perpendicular bisectors of the sides of the triangle.*
- D. *the triangle altitudes (or their extensions).*

Question 458 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Through two points, we can draw one line. Through three points, three lines can be drawn. Through thirty points, how many lines can we draw?

- A. 435 lines.
- B. 445 lines.
- C. 455 lines.
- D. 465 lines.

Question 460 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

A polyhedron with 20 faces and 30 edges has:

- A. 12 vertices.
- B. 6 vertices.
- C. 18 vertices.
- D. 24 vertices.

Question 461 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Which of the following sentences has only platonic solids?

- A. *Tetrahedron, octahedron, and icosahedron.*
- B. *Hexahedron, decahedron, and dodecahedron.*
- C. *Tetrahedron, heptahedron, and icosahedron.*
- D. *Cone, cylinder, and sphere.*

Question 462 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Select the platonic solid whose sum of the number of faces with the number of vertices is 14.

- A. *Octahedron.*
- B. *Tetrahedron.*
- C. *Dodecahedron.*
- D. *Icosahedron.*

Question 463 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

The sum of the number of faces with the number of vertices of a solid is 17. Choose the correct answer.

- A. *The solid is a pentagonal prism.*
- B. *The solid is a hexagonal prism.*
- C. *The solid is a hexagonal pyramid.*
- D. *The solid is a pentagonal pyramid.*

Question 464 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

The number of diagonals of an n -sided polygon is

- A. $\frac{n(n-3)}{2}$.
- B. $n(n-3)$.
- C. $2n-2$.
- D. $2n(n-3)$.

Question 467 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Choose the true statement.

- A. *A concave polygon may have a diagonal outside the polygon.*
- B. *A convex polygon may have a diagonal outside the polygon.*
- C. *The diagonals of a convex polygon are all congruent.*
- D. *A diagonal can join two consecutive vertices of the polygon.*

Question 493 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Which set of numbers stands for the lengths of the sides of a triangle?

- A. $\{2, 4, 5\}$.
- B. $\{3, 3, 6\}$.
- C. $\{2, 4, 8\}$.
- D. $\{1, 3, 7\}$.

Question 494 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

The amplitude of an exterior angle of a regular hexagon is

- A. 60° .
- B. 120° .
- C. 30° .
- D. 90° .

Question 495 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

The amplitude of an interior angle of a regular octagon is

- A. 135° .
- B. 125° .
- C. 140° .
- D. 130° .

Question 496 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Choose the value that corresponds to the sum of the amplitudes of the internal angles of an octagon.

- A. 1080° .
- B. 900° .
- C. 360° .
- D. 1440° .

Question 497 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Knowing that P is a parallelogram, indicate what you can say about P .

- A. The sum of the amplitudes of the internal angles of P is 360° .
- B. P has four acute angles.
- C. P has four right angles.
- D. P has two congruent diagonals.

Question 498 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Choose the true statement.

- A. *Ascalenetrianglecanhavethreeacuteangles.*
- B. *Thereisanequilateraltrianglewitharightangle.*
- C. *Ascalenetrianglecanhavetwocongruentangles.*
- D. *Anisoscelestrianglecanhavethreegeometricallydifferentangles.*

Question 499 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Consider that T is a right-angled triangle. Choose the true statement.

- A. T has three sides with lengths $4cm$, $3cm$ and $5cm$.
- B. T has three sides with lengths $4cm$, $3cm$ and $6cm$.
- C. T may have an obtuse angle.
- D. The amplitude of the other angles of T is 45° .

Question 500 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Choose the true statement.

- A. If T is an equilateral triangle, then T is an acute triangle.
- B. If T is an acute triangle, then T is an equilateral triangle.
- C. If T is a right triangle, then T is a scalene triangle.
- D. If T is an obtuse triangle, then T is an isosceles triangle.

Question 501 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

The total area (expressed in a given measure unit) of this cylinder is:

- A. $2\pi a(a + b)$.
- B. $2\pi b(a + b)$.
- C. $\pi a(2a + b)$.
- D. $\pi a(a + 2b)$.

Question 502 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Consider a dilation (an isotropic scaling) of centre O and scale factor r . Choose the correct statement.

- A. *The original figure and its transformed one are always similar.*
- B. If $r = 2$ the dilation corresponds to a translation.
- C. If $r > 1$ the dilation is an isometry.
- D. *The original figure and its transformed one are always congruent.*

Question 503 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Indicate which geometric transformation always preserves the direction and orientation of directed line segments.

- A. *Translation.*
- B. *Rotation.*
- C. *Reflection.*
- D. *Gliding reflection.*

Question 504 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Indicate which of the following geometrical transformations is not an isometry.

- A. Dilation (isotropic scaling) of centre $(0, 0)$ and scale factor 2.
- B. Translation associated to vector $\vec{u} = (2, 3)$.
- C. Rotation of centre $(0, 0)$ and amplitude (angle) 90° .
- D. Line of reflection $y = x$.

Question 505 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Considering the similarity criteria of triangles, choose the correct statement.

- A. *Two scalene triangles are similar if two internal angles of one of them are congruent to two of the internal angles of the other.*
- B. *Two isosceles triangles are similar if the length of two sides of one of them is directly proportional to the length of two sides of the other.*
- C. *Two right triangles are always similar.*
- D. *Two scalene triangles are never similar.*

Question 506 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Find the correct relationship between the type of triangle and the number of lines of symmetry.

- A. *Anequilateraltrianglehas3linesofsymmetry.*
- B. *Anisoscelestrianglehas2linesofsymmetry.*
- C. *Ascalenetrianglehas1lineofsymmetry.*
- D. *Anequilateraltrianglehas6linesofsymmetry.*

Question 507 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

A square allows for:

- A. *4reflectionalsymmetriesand4rotationalsymmetries.*
- B. *4reflectionalsymmetriesand8rotationalsymmetries.*
- C. *8reflectionalsymmetriesand4rotationalsymmetries.*
- D. *8reflectionalsymmetriesand8rotationalsymmetries.*

Question 508 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

A regular pentagon allows for five rotational symmetries. Which are the amplitudes of such rotational symmetries:

- A. 72° , 144° , 216° , 288° and 360° .
- B. 36° , 72° , 108° , 144° and 180° .
- C. 36° , 72° , 144° , 288° , 360° .
- D. 90° , 180° , 270° , 360° , 450° .

Question 509 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

A regular hexagon allows for:

- A. 6 rotational symmetries.
- B. 3 reflectional symmetries.
- C. 9 reflectional symmetries.
- D. 12 rotational symmetries.

Question 510 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Choose the incorrect statement.

- A. *Reflection is the sole isometry that does not preserve the correct direction.*
- B. *An isometry is a geometrical transformation in which the original figure and its transformed are congruent.*
- C. *A transformation is a similarity.*
- D. *Rotation and translation are isometries that preserve the orientation of the angles.*

Question 511 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Consider a right-angled triangle, where the hypotenuse and one side measure 10 cm and 6 cm in length, respectively. Indicate the length of the other side.

- A. *8cm*
- B. *7cm*
- C. *3cm*
- D. *4cm*

Question 512 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Two triangles are congruent if, from one to the other, they have:

- A. *A congruent side and both adjacent angles to that side also congruent.*
- B. *Two congruent sides.*
- C. *Two congruent angles.*
- D. *Two congruent sides. And an angle which is also congruent.*

Question 513 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Imagine a triangle whose sides are of length 8cm, 6cm and 3cm:

- A. *The triangle is scalene.*
- B. *The triangle is isosceles.*
- C. *The triangle is acute – angled.*
- D. *The triangle only has an angle with amplitude less than 90° .*

Question 514 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Imagine a triangle whose amplitude of two of the angles is 70° and 30° :

- A. *The triangle is acute – angled.*
- B. *The triangle is straight.*
- C. *The triangle is obtuse – angled.*
- D. *The triangle is isosceles.*

Question 515 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

In quadrilateral $[ABCD]$ the diagonals are congruent, orthogonal, they are bisectors of the internal angles and intersect in the median point which is a symmetry center. $[ABCD]$ is a:

- A. *Square.*
- B. *Rhombus.*
- C. *Rectangle.*
- D. *Oblique parallelogram.*

Question 516 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

In a rectangle:

- A. *There are two lines of symmetry and their diagonals are congruent and intersect in the median point which is a symmetry center.*
- B. *There is just one line of symmetry and its diagonals are congruent and intersect in the median point which is a symmetry center.*
- C. *There are two lines of symmetry and their diagonals are congruent and intersect in the median point.*
- D. *There are four lines of symmetry and their diagonals are congruent and intersect in the median point which is a symmetry center.*

Question 517 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

In an isosceles trapezoid:

- A. *The straight line which crosses the median points is a line of symmetry.*
- B. *The sides are congruent two by two.*
- C. *There are two lines of symmetry.*
- D. *Diagonals bisect.*

Question 518 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Mark the true statement.

- A. The sum of the amplitudes of the external angles of any triangle is always equal to 360° .
- B. The sum of the amplitudes of the external angles of a convex quadrilateral is always equal to 540° .
- C. The sum of the amplitudes of the external angles of a convex pentagon is always equal to 720° .
- D. The sum of the amplitudes of the external angles of a convex hexagon is always equal to 900° .

Question 519 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

The sum of the amplitudes of the internal angles of a convex polygon with five sides is:

- A. 540° .
- B. 360° .
- C. 900° .
- D. 720° .

Question 520 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Choose the correct statement.

- A. The sum of the amplitudes of the external angles of any convex polygon is 360° .
- B. The sum of the external angles of any convex polygon is 180° .
- C. The sum of the external angles of any convex polygon is 720° .
- D. The sum of the external angles of any convex polygon is $(n - 2) \times 180^\circ$, where n is the number of sides of the polygon.

Question 521 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

A polygon is regular when:

- A. *All its sides have the same length and all angles have the same amplitude.*
- B. *The parallel sides have the same length and opposite angles have the same amplitude.*
- C. *All the angles have the same amplitude.*
- D. *All the sides have the same length.*

Question 522 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Each external angle of a given polygon has 60° amplitude. Which is the sum of the amplitudes of the internal angles of this polygon?

- A. 720°
- B. 180°
- C. 540°
- D. 360°

Question 523 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

The diagonal of a polygon is:

- A. *A straight line segment whose endpoints are two non – consecutive vertices.*
- B. *A straight line segment uniting the median point of two consecutive sides.*
- C. *A straight line through two non – consecutive vertices.*
- D. *A semi – straight line whose origin is one of the vertices of the polygon and goes through the opposite vertex.*

Question 524 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

In a cartesian referential two consecutive vertices A and B , of a parallelogram $[ABCD]$ were marked, together with point M corresponding to the intersection of its diagonals. Knowing that the coordinates of points A , B and M are $(-1, -3)$, $(3, -3)$ and $(2, 0)$, respectively, the coordinates of points C and D are:

- A. $C(5, 3)$ and $D(1, 3)$
- B. $C(5, 2)$ and $D(3, 2)$
- C. $C(2, 3)$ and $D(1, 3)$
- D. $C(5, 3)$ and $D(3, 2)$

Question 1256 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

If a triangle has a square angle and the two adjacent sides are equal in length, what are the other two angles in the triangle?

- A. 45degrees each
- B. $90\text{degrees and }0\text{degrees}$
- C. 33degrees each
- D. 90degrees each

Question 1257 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

How many sides does a square have?

- A. 4
- B. 3
- C. 2
- D. 12

Question 1645 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Which are the polygons? Mark the correct answer.

- A. B, D, E, F and G .
- B. D, E and G .
- C. A, D, E, G and H .
- D. B, C and F .

Question 1782 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

T is an isosceles right triangle. Select the option that corresponds to the length of the sides of T.

- A. $2cm, 2cm, 2\sqrt{2}cm$.
- B. $2cm, 2cm, 4cm$.
- C. $2cm, 2cm, 8cm$.
- D. $2cm, 2cm, \sqrt{2}cm$.

Question 1785 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

The sum of the number of faces with the number of vertices of a solid is 20. Choose the right answer.

- A. The solid is a enneagonal pyramid.
- B. The solid is a decagonal pyramid.
- C. The solid is a decagonal prism.
- D. The solid is a enneagonal prism.

Question 1794 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Select the correct statement.

- A. If a polygon is regular, then their diagonals are congruent.
- B. If a polygon has two pairs of parallel sides, then it is a parallelogram.
- C. If a polygon has four right angles, then it is a rectangle.
- D. If a polygon is not regular, then it doesn't have reflection symmetries.

Question 1797 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

How many diagonals can we trace on a decagon?

- A. 35
- B. 20
- C. 23
- D. 12

Question 1799 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Indicate which of the following geometrical transformations is not an isometry.

- A. Homotetia of centre $(0, 0)$ and ratio 4.
- B. Homotetia of centre $(-2, 2)$ and ratio -1 .
- C. Rotation of centre $(-2, 2)$ and amplitude 45° .
- D. Rotation of centre $(0, 0)$ and amplitude 180° .

Question 1809 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Select the wrong statement.

- A. *The lateral faces of a quadrangular prism are always perpendicular to the bases.*
- B. *All the opposite faces of any parallelepiped are parallel.*
- C. *The lateral faces of all prisms are parallelograms.*
- D. *All cubes are prisms.*

Question 1814 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Select the wrong statement.

- A. *The base of a pyramid must be a convex polygon.*
- B. *The base of a pyramid can be any polygon.*
- C. *The apex of a pyramid is the intersection point of all the lateral edges.*
- D. *The lateral faces of any pyramid are triangles.*

Question 1820 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Select the true statement.

- A. *Prisms and pyramids can be non-convex polyhedrons.*
- B. *All pyramids with three lateral faces are platonic solids.*
- C. *Prisms and pyramids are regular polyhedrons.*
- D. *The bases of prisms and pyramids are always regular polygons.*

Question 1831 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Choose the wrong statement.

- A. *In a translation, the orientation of the angle is inverted.*
- B. *Translation, rotations, reflections and sliding reflections are isometries.*
- C. *In a sliding reflection the orientation of the angle is inverted.*
- D. *There are homotheties that preserve the orientation of directed line segments.*

Question 2107 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

A triangle has no diagonals. A quadrilateral has two diagonals. A pentagon has five diagonals. How many diagonals a fifty side polygon have?

- A. *1175 diagonals*
- B. *1075 diagonals*
- C. *1125 diagonals*
- D. *1225 diagonals*

Question 2109 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Consider that to a square of side size 4cm one applies a centre homotetia isotropic scaling on the square's vertexes and ratio r . Select the wrong statement.

- A. *If $r = 3$ the area of the resulting square is triple of the starting square.*
- B. *If $r > 1$ the perimeter of the resulting square is r times bigger than the perimeter of the original square*
- C. *If $|r| = 1$ the homotetia is an isometry*
- D. *If $|r| > 1$ the length of the size of the transformed square is $4|r|$ cm.*

Question 2111 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Select the true statement.

- A. *In the sliding reflection the translation vector has the direction of the reflection axis.*
- B. *Translation preserves the direction of straight line segments but cannot preserve it's orientation.*
- C. *Rotation and reflection may not be isometries.*
- D. *An isometry always respects the orientation of the angles.*

Question 2113 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

A square allows for four rotation symmetries. What are the amplitudes of such rotation symmetries:

- A. 90° , 180° , 270° and 360° .
- B. 45° , 90° , 135° and 180° .
- C. 45° , 90° , 135° and 180° .
- D. 90° , 180° , 360° and 720° .

Question 2115 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

A regular pentagon allows for:

- A. *5 reflections symmetries and 5 rotations symmetries.*
- B. *5 reflections symmetries and 10 rotations symmetries.*
- C. *10 reflections symmetries and 5 rotations symmetries.*
- D. *10 reflections symmetries and 10 rotations symmetries.*

Question 2117 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Consider the ABC triangle whose perimeter is 39 units. Knowing that $\overline{AB} = 3x - 2$, $\overline{BC} = 4x + 1$ and $\overline{AC} = 2x + 4$, state which of the triangles angles has the largest amplitude.

- A. $\angle CAB$.
- B. $\angle ABC$.
- C. $\angle BCA$.
- D. *All angles have the same amplitude.*

Question 2119 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Mark the wrong statement:

- A. The sum of the amplitudes of the external angles of a triangle always equals 180° .
- B. The sum of the amplitudes of the internal angles of a triangle always equals 180° .
- C. *The amplitude of each external angle of a triangle equals the sum of the amplitudes of both nonadjacent internal angles.*
- D. *In a triangle the longest side is always opposite to the angle with the biggest amplitude.*

Question 2121 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

In a quadrilateral $[ABCD]$ the diagonals are orthogonal, they bisect the internal angles, they are the only symmetry axis and intersect in the median point which is a symmetry center. $[ABCD]$ is a:

- A. *Rhombus.*
- B. *Square.*
- C. *Rectangle.*
- D. *Oblique parallelogram.*

Question 2123 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

The sum of the internal angles (S_i) of any convex polygon is:

- A. $S_i = (n - 2) \times 180^\circ$, where n is the number of sides of the polygon.
- B. $S_i = n \times 180^\circ$, where n is the number of sides of the polygon.
- C. $S_i = (n + 2) \times 180^\circ$, where n is the number of sides of the polygon.
- D. $S_i = (n - 2) \times 360^\circ$, where n is the number of sides of the polygon.

Question 2125 (Fundamental Mathematics – Elementary Geometry | LLM difficulty:)

Choose the correct statement.

- A. An external angle of a regular polygon can measure 15° .
- B. There is a five-sided convex polygon whose internal angles have amplitudes equal to 95° , 100° , 115° , 70° and 120° .
- C. An internal angle of a regular polygon can measure 50° .
- D. *If a given polygon is regular. The external angles may not all have the same amplitude.*